# Utility aware Gaussian Process Mapping

Bowen Tan

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## 1 Introduction

This report details the experimental implementation and results of a utility-aware privacy protection scheme for Gaussian Process (GP) spatial data. The goal is to maximize privacy in sensitive regions while guaranteeing a minimum level of utility in task-relevant regions.

## 2 Experimental Scheme

### 2.1 Data Generation and Region Selection

We generated $n = 50$ random coordinates within a $[0, 1]^2$ domain. The spatial correlation is modeled using an RBF kernel with a lengthscale of 0.2. The observation noise variance is set to $v = 10^{-2}$.

To ground the mathematical formulation in a physical spatial context, we explicitly define the sensitive region $\mathcal{S}$ and the task-relevant region $\mathcal{T}$ within the $[0, 1]^2$ domain instead of assigning arbitrary mathematical properties.

**1. Region Definition:**

- **Sensitive Region ($\mathcal{S}$):** We define a localized rectangular area $\mathcal{S} = [0.4, 0.6] \times [0.4, 0.6]$ to represent a highly restricted physical location (e.g., a military facility). We sample $N_S = 10$ coordinates uniformly within this bounding box.
- **Task Region ($\mathcal{T}$):** The task region represents the global utility requirement (e.g., general road navigation). We define $\mathcal{T}$ as a uniform $10 \times 10$ grid covering the entire $[0, 1]^2$ domain, yielding $N_T = 100$ spatial coordinates.

**2. Matrix Generation Algorithm:** Based on the Gaussian Process covariance structure, the information accuracy corresponding to a spatial region is governed by the cross-covariance between the observation points $X$ ($n = 50$) and the target regions. Let $K(\cdot, \cdot)$ denote the RBF kernel matrix. The weighting matrices $M_S$ and $M_T$ are deterministically generated using the following algebraic mapping:

$$M_S = K(X, \mathcal{S})K(\mathcal{S}, X) \in \mathbb{R}^{n\times n} \tag{1}$$

$$M_T = K(X, \mathcal{T})K(\mathcal{T}, X) \in \mathbb{R}^{n\times n} \tag{2}$$

*Remark on Numerical Properties:* Because $M_S$ is constructed via the inner product of $K(X, \mathcal{S})$ which has $N_S = 10$ columns, the resulting matrix $M_S$ naturally forms a low-rank Positive Semi-Definite (PSD) matrix with $\text{rank}(M_S) \leq 10$. This provides a rigorous physical justification for the low-rank numerical settings investigated in earlier sections. Conversely, because $\mathcal{T}$ densely covers the domain with $N_T = 100 > n$, $M_T$ inherently becomes a full-rank PSD matrix.

### 2.2 Optimization Problem

For GPR with an additive synthetic noise covariance matrix $C$, the posterior covariance over the sensitive region $S$ is

$$\text{Cov}(S) = K(S,S) - K(S,X)(K(X,X) + V + C)^{-1}K(X,S).$$

Let

$$A = K(X,X) + V, \qquad W = (A + C)^{-1}.$$

Then the posterior covariance can be written as

$$\text{Cov}(S) = K(S,S) - K(S,X)WK(X,S).$$

After solving the optimization problem for $W^*$, the corresponding physical synthetic noise covariance is reconstructed as

$$C_{\text{opt}} = (W^*)^{-1} - A.$$

Therefore, the optimized matrix $W^*$ is not the noise covariance itself, but the inverse of the noise-augmented observation covariance. Minimizing privacy leakage is to maximize $Cov(S)$ (in the sense of maximizing $\text{Tr}(Cov(S))$), that is to minimize (using $\text{Tr}(AB) = \text{Tr}(BA)$)

$$\text{Tr}(K(S,X)WK(X,S)) = \text{Tr}(K(X,S)K(S,X)W) =: \text{Tr}(M_S W).$$

**Practical Interpretation of $\lambda$ and $\eta$**

The two tuning parameters play different roles in the proposed optimization problem. The parameter $\lambda > 0$ is a regularization parameter. It is not used to enforce privacy or utility directly; rather, it stabilizes the optimization problem and helps select a unique and well-conditioned optimizer. In practice, $\lambda$ can be chosen through a small logarithmic sweep, and we select the smallest value that yields stable numerical behavior without significantly changing the achieved utility or privacy leakage.

The parameter $\eta$ is used in the current full-rank numerical implementation. By imposing

$$W \succeq \eta I, \qquad \eta > 0,$$

the optimized matrix $W$ is guaranteed to be invertible, so that the physical synthetic noise covariance can be reconstructed as

$$C_{\text{opt}} = (W^*)^{-1} - A.$$

Therefore, $\eta$ should be interpreted as a numerical lower bound for full covariance reconstruction and Gaussian noise sampling in the original observation space, rather than as a fundamental privacy parameter. Theoretically, the privacy mechanism does not necessarily require $\eta > 0$. If $\eta = 0$, the optimizer $W$ may be singular, in which case the mechanism can be interpreted through a lower-dimensional subspace-sharing formulation.

The privacy protection is formulated as a Semi-Definite Program (SDP), Where $A = K(X,X) + V$ is the noisy covariance matrix of observations:

$$\begin{aligned}
\min_{W} \quad & \text{Tr}(M_S W) + \lambda \|W\|_F^2 \\
\text{s.t.} \quad & W \succeq \eta I, \\
& A^{-1} - W \succeq 0, \\
& \text{Tr}(M_T W) \geq \epsilon',
\end{aligned}$$

where $M_S$ represents the sensitive-region weighting matrix, $M_T$ represents the task-region weighting matrix, and $\epsilon'$ is the required utility threshold. Since

$$W = (A + C)^{-1},$$

a larger value of $\mathrm{Tr}(M_T W)$ indicates that more predictive information is retained in the task-relevant region $T$. Therefore, the constraint

$$\mathrm{Tr}(M_T W) \geq \epsilon'$$

ensures that the injected synthetic noise does not destroy the minimum prediction utility required for the target task.

### 2.3 Uniqueness and Full-Rank Implementation

**Theorem 1** (Uniqueness of Solution)**.** *Consider the feasible set*

$$\mathcal{F} = \left\{ W \in \mathbb{S}^n \;\middle|\; W \succeq \eta I,\ A^{-1} - W \succeq 0,\ \mathrm{Tr}(M_T W) \geq \epsilon' \right\}.$$

*Assume that the feasible set $\mathcal{F}$ is nonempty. If the objective function is regularized as*

$$f(W) = \mathrm{Tr}(M_S W) + \lambda \|W\|_F^2,$$

*then for any $\lambda > 0$, the proposed SDP problem admits a unique global optimizer. When $\lambda = 0$, the objective becomes linear, and uniqueness is not guaranteed.*

*Proof.* The feasible set $\mathcal{F}$ is convex because it is defined by the intersection of convex constraints. Specifically, the constraint

$$W - \eta I \succeq 0$$

is a shifted positive semidefinite constraint, the constraint

$$A^{-1} - W \succeq 0$$

is also a positive semidefinite constraint, and

$$\mathrm{Tr}(M_T W) \geq \epsilon'$$

is an affine inequality.

When $\lambda > 0$, the term $\|W\|_F^2 = \mathrm{Tr}(W^\top W)$ is strictly convex over $\mathbb{S}^n$. Since $\mathrm{Tr}(M_S W)$ is linear in $W$, the objective function

$$f(W) = \mathrm{Tr}(M_S W) + \lambda \|W\|_F^2$$

is strictly convex for any $\lambda > 0$. A strictly convex function minimized over a convex feasible set admits at most one global minimizer. Therefore, the optimal solution $W^*$ is unique.

When $\lambda = 0$, the objective reduces to the linear function

$$f(W) = \mathrm{Tr}(M_S W).$$

A linear objective over a convex feasible set does not, in general, guarantee a unique optimizer. In particular, suppose there exists an optimal solution $W^* \in \mathcal{F}$ that lies in the relative interior of the feasible set and there exists a nonzero perturbation direction $\Delta \in \mathbb{S}^n$, $\Delta \neq 0$, satisfying

$$\mathrm{Tr}(M_S \Delta) = 0, \qquad \mathrm{Tr}(M_T \Delta) = 0.$$

Because $W^*$ is assumed to lie in the relative interior of $\mathcal{F}$, there exists a sufficiently small scalar $\delta \neq 0$ such that

$$W^* + \delta\Delta \in \mathcal{F}.$$

The utility constraint is preserved since

$$\mathrm{Tr}(M_T(W^* + \delta\Delta)) = \mathrm{Tr}(M_T W^*) + \delta\mathrm{Tr}(M_T\Delta) = \mathrm{Tr}(M_T W^*) \geq \epsilon'.$$

The objective value is also preserved:

$$f(W^* + \delta\Delta) = \mathrm{Tr}(M_S(W^* + \delta\Delta)) = \mathrm{Tr}(M_S W^*) + \delta\mathrm{Tr}(M_S\Delta) = f(W^*).$$

Thus $W^* + \delta\Delta$ is another feasible solution with the same optimal objective value. Since $\delta\Delta \neq 0$, this solution is distinct from $W^*$. Hence, when $\lambda = 0$, the optimizer may be non-unique. Therefore, the squared Frobenius regularization with $\lambda > 0$ provides a sufficient condition for uniqueness. □

**Proposition 2 (Full-Rank Numerical Implementation)**

If the feasible set enforces

$$W \succeq \eta I, \quad \eta > 0,$$

then every feasible solution, including the optimal solution $W^*$, is invertible.

*Proof.* Since $W^* \succeq \eta I$, every eigenvalue of $W^*$ satisfies

$$\lambda_i(W^*) \geq \eta > 0.$$

Therefore,

$$\det(W^*) = \prod_{i=1}^{n} \lambda_i(W^*) > 0.$$

Hence $W^*$ is nonsingular and $(W^*)^{-1}$ exists.

This proposition justifies the full-rank numerical implementation used in the current experiments. It should not be interpreted as a theoretical necessity of the privacy mechanism itself. If $\eta = 0$, a singular optimizer $W$ may still be interpreted through a subspace-sharing mechanism, which requires separate verification of the induced covariance and utility constraint. □

## 2.4 Singular-$W$ Subspace Mechanism

The full-rank implementation in the previous subsection relies on the constraint

$$W \succeq \eta I, \qquad \eta > 0,$$

so that $W$ is invertible and the physical synthetic noise covariance can be reconstructed as

$$C_{\mathrm{opt}} = (W^*)^{-1} - A.$$

However, this full-rank reconstruction is not theoretically necessary for the privacy mechanism. If we set $\eta = 0$, the SDP only requires

$$W \succeq 0,$$

and the optimizer $W^*$ may be singular. In this case, $(W^*)^{-1}$ does not exist as a finite covariance matrix over the full observation space. Therefore, the singular case should not be interpreted as a full-space additive-noise mechanism. Instead, it can be interpreted as a subspace-release mechanism.

Suppose a spectral decomposition of the singular optimizer $W^*$ is

$$W^* = \begin{pmatrix} O_1 & O_2 \end{pmatrix} \begin{pmatrix} \Lambda & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} O_1^T \\ O_2^T \end{pmatrix} = O_1 \Lambda O_1^T,$$

where $\Lambda$ contains the positive eigenvalues of $W^*$, and the columns of $O_1$ span the finite-information subspace. The orthogonal complement $O_2$ corresponds to the null space of $W^*$.

Under this interpretation, the data owner releases only the projected component

$$Y_1 = O_1^T Y,$$

and does not release

$$Y_2 = O_2^T Y.$$

The unreleased component $Y_2$ is equivalently protected by infinite synthetic noise, or zero released precision. Therefore, the effective precision matrix of the released information is still

$$W^* = O_1 \Lambda O_1^T.$$

To implement the finite shared component, we add Gaussian noise only in the $O_1$-subspace. Since the covariance of the projected observation $O_1^T Y$ is

$$O_1^T A O_1,$$

and the target covariance corresponding to precision $\Lambda$ is

$$\Lambda^{-1},$$

the required subspace synthetic noise covariance is

$$C_1 = \Lambda^{-1} - O_1^T A O_1.$$

Thus the released noisy subspace observation is

$$Z_1 = O_1^T Y + \xi_1, \qquad \xi_1 \sim \mathcal{N}(0, C_1).$$

For this singular mechanism to be physically valid, the induced subspace covariance must satisfy

$$C_1 \succeq 0.$$

If this condition holds, then $Z_1$ has covariance

$$\mathrm{Cov}(Z_1) = O_1^T A O_1 + C_1 = \Lambda^{-1},$$

so that its precision is exactly $\Lambda$. Consequently, the full-space effective precision matrix induced by the released data is

$$O_1 \Lambda O_1^T = W^*.$$

The privacy and utility constraints can therefore still be evaluated using the same effective precision matrix $W^*$. In particular, the utility requirement remains

$$\mathrm{Tr}(M_T W^*) \geq \epsilon',$$

while the sensitive-region privacy leakage remains

$$\mathrm{Tr}(M_S W^*).$$

The difference from the full-rank implementation is that the mechanism no longer reconstructs a finite full-space covariance matrix $C_{\mathrm{opt}}$. Instead, it releases only the finite-precision subspace and withholds the null-space component. In the current experiments, we use the full-rank formulation with $\eta > 0$ for direct covariance reconstruction and Gaussian sampling. The singular formulation above provides a valid extension when the PSD condition

$$\Lambda^{-1} - O_1^T A O_1 \succeq 0$$

is satisfied and the utility constraint is rechecked under the projected release mechanism.

## 3 Results and Analysis

The experiments were conducted using the CVXPY interface with the optimal solver. The spatial parameters were physically mapped rather than arbitrarily assigned, yielding a mathematically consistent evaluation. The numerical results are summarized below.

**Full-rank implementation.** We first evaluate the full-rank implementation with $\eta > 0$. In this case, $W^*$ is invertible and the physical synthetic noise covariance can be reconstructed as

$$C_{\text{opt}} = (W^*)^{-1} - A.$$

The full-rank numerical results are:

- **Solver Status:** Optimal
- **Target Utility** ($\epsilon'$)**:** 10.0
- **Achieved Utility:** 10.0008
- **Optimal Privacy Leakage:** 0.0065
- **Noise Matrix Stability:** Since the full-rank implementation uses $\eta > 0$, the inverse $(W^*)^{-1}$ is well-defined. After reconstructing

  $$C_{\text{opt}} = (W^*)^{-1} - A,$$

  we check its eigenvalues and apply PSD projection when necessary. The minimum eigenvalue of the final covariance matrix is 0.508, confirming that it is numerically positive semidefinite and suitable for Gaussian noise sampling.

**Singular-$W$ subspace verification.** We further evaluate the singular formulation by setting $\eta = 0$. In this case, the SDP does not enforce a full-rank lower bound on $W$, so the optimizer is allowed to be positive semidefinite and singular. The solver returns an optimal solution with

$$\text{rank}(W^*) = 44 < 50.$$

Thus, the optimized mechanism releases only a 44-dimensional finite-information subspace and withholds the remaining null-space component.

The achieved task utility is

$$\text{Tr}(M_T W^*) = 10.0019 \geq \epsilon' = 10.0,$$

so the required utility constraint remains satisfied.

$$\text{Tr}(M_S W^*) = 8.45 \times 10^{-8},$$

which is nearly zero.

To verify that the singular solution can be implemented as a valid subspace Gaussian mechanism, we decompose

$$W^* = O_1 \Lambda O_1^T$$

over its positive-eigenvalue subspace and compute the induced subspace synthetic noise covariance

$$C_1 = \Lambda^{-1} - O_1^T A O_1.$$

The minimum eigenvalue of $C_1$ is

$$\lambda_{\min}(C_1) = 0.4977 > 0,$$

confirming that

$$C_1 \succeq 0.$$

Therefore, the singular-$W$ solution is physically valid as a subspace-release mechanism: the finite-information component is released with Gaussian noise covariance $C_1$, while the null-space component is not released and is equivalently protected by infinite variance.

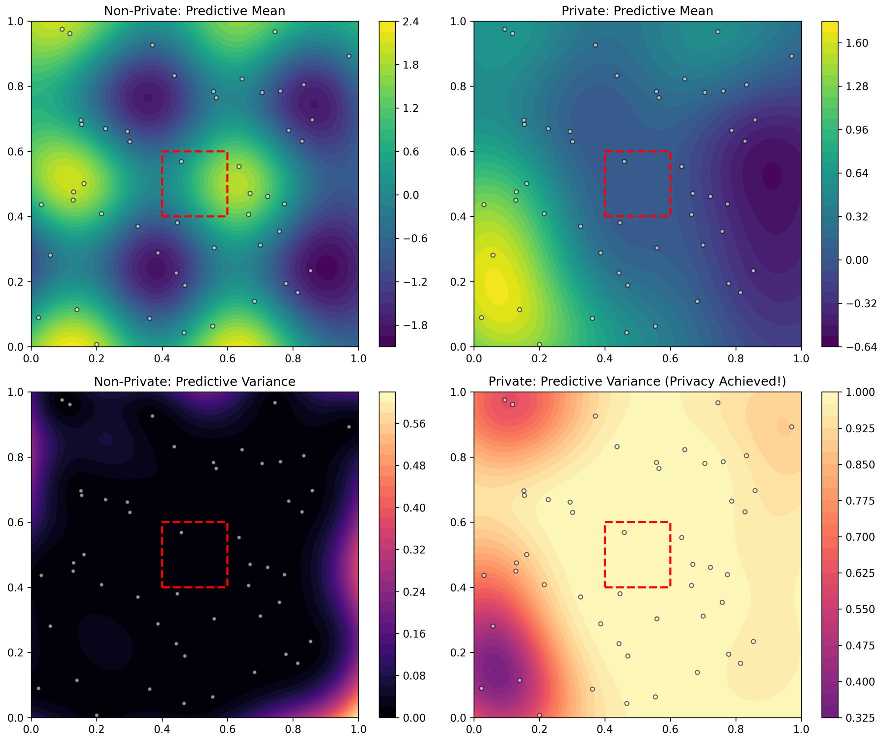


Figure 1: GPR predictive performance comparison between the non-private baseline (left) and the proposed utility-aware private framework (right). The red dashed box encapsulates the designated sensitive region $\mathcal{S}$.

As illustrated in Figure 1, the full-rank implementation exhibits the expected qualitative behavior:

- **Utility Preservation in $\mathcal{T}$:** The predictive mean and low-variance structural distribution outside the sensitive region remain congruent with the non-private baseline (e.g., the high-utility spatial peaks in the lower-left domain are perfectly preserved).
- **Obfuscation in $\mathcal{S}$:** Within the restricted spatial boundaries of $\mathcal{S}$ (red dashed box), the true multi-modal topography of the predictive mean is significantly flattened and suppressed. Concurrently, the predictive variance undergoes a substantial, systemic escalation (shifting from near-zero uncertainty up to the maximum theoretical bound of 1.0).

This empirical result supports the claim that, in the full-rank implementation, the synthesized correlated noise

$$\xi \sim \mathcal{N}(0, C_{\text{opt}})$$

can selectively reduce high-confidence estimation in localized sensitive regions while maintaining the prescribed level of spatial prediction utility for the broader task domain.

## 4 Conclusion

In conclusion, the empirical results suggest that the proposed framework can achieve a controlled trade-off between task utility and localized privacy in Gaussian Process spatial mapping. Specifically, the semidefinite programming (SDP) optimization strictly satisfies the designated target utility threshold ($\epsilon' = 10.0$) while compressing the optimal privacy leakage to a minimal bound of 0.0065. Furthermore, the reconstructed physical noise covariance matrix

$$C_{\text{opt}} = (W^*)^{-1} - A$$

has a minimum eigenvalue of 0.508 after projection, confirming that the final covariance matrix is numerically positive semidefinite and suitable for Gaussian noise sampling. This framework provides a practical and mathematically grounded full-rank implementation for utility-aware privacy-preserving Gaussian Process spatial mapping. In the current implementation, $\eta > 0$ is used as a numerical lower bound for covariance reconstruction and Gaussian noise sampling, rather than as a fundamental privacy parameter. A more general singular-$W$ formulation can be interpreted through subspace sharing, where only the range-space component is released and the null-space component is protected by infinite variance. The singular-$W$ formulation is also numerically verified. When $\eta = 0$, the optimizer has rank $44 < 50$, and the induced subspace covariance

$$C_1 = \Lambda^{-1} - O_1^T A O_1$$

has minimum eigenvalue $0.4977 > 0$, confirming that the subspace Gaussian mechanism is physically valid in this experiment. The utility constraint remains satisfied with

$$\text{Tr}(M_T W^*) = 10.0019 \geq 10.0.$$

This shows that $\eta > 0$ is not theoretically necessary; it is mainly needed for the full-rank covariance reconstruction implementation.